A Sovereign Button Detection and Measure the Alignment Using Image Processing

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***Abstract*— Modern competitive world is combined with various industries. Among them, the apparel industry takes a prominent place. When considering the process of apparel industry, there can be seen a huge process from material purchasing to the manufacturing of the end products. Therefore, Seamstress needs to maintain efficiency and effectiveness properly through-out the entire sewing process. Somehow, Defining dimensions and accuracy is difficult to maintain by humans without getting involved in technology. Correct dimensions can be taken as the highest need when doing the apparel productions. But, for some instant seamstresses did not tend to measure lengths with the use of correct attributes. In that respect, a novel system can be introduced through this research which will be capable to monitor the measuring lengths and detect measurement errors. Moreover, product quality can be led to its demanding, it fulfils the customer satisfaction and enhance standard of the product. Whether the defects and the quality being maintained, it can be helped to improve the productivity. The goal of this research is to identify the measurement defects that involves to reduce the quality and productivity of men's' plain collared shirts by controlling imperfections and measurement errors. Moreover, if the system is able to identify the measurement errors accurately, then the system will be able to define the correct measurements and finalized the product status as a quality product. By identifying defects, it is possible to minimize the major faults and minor production losses. Defects in the apparel industry can be threatened adversely to the economical factor as well.**

Keywords— Interpretation, Detection, Measurement, Alignment, Recognition

I. Introduction

When considering the apparel industry, the problem has been encountered with the sewing process of men's shirts. The major problem is shirt buttons are not aligned properly. Therefore, the output of the end product is coming without quality. Through this research, emphasis detecting defects and finding out the solution. The reason to do this research, the minor sewing defects may lead to diminishing quality of the entire production. This has introduced a better solution for sustaining the quality of the product to meet consumer needs. In addition, consideration of environmental friendliness is another essential factor to gain when providing a solution to the apparel industry.

As a consequence of pursuing the quality and performance of sewing men's shirts, the solution can be provided to the alignment issues by measuring the button length starting from the top button to the next nearest button. According to researches, being able to find the length from the top button to the next nearest button should be kept 2 inches (2") and this will be a slightly standard measurement. Human power is required to control sewing machines. [1]. Further, human power and production of shirts can be enhanced by introducing a better solution to the shirt button alignment issue by setting the button alignment between the first two buttons as 2 inches.

II. Research Background

A. Problem in Brief

According to the gathered details from the apparel industry, able to identify one of the prominent inefficiency that has been occurred when sewing men's shirts. [2] To emphasis the research area, able to add limitations with choosing the errors which can be occurred when measuring the button alignment of the shirt. Most of the men came across for inconveniences due to the button line of their shirts being unaligned. Due to that reason, when men are wearing the shirt, it could be clumsy. A seamstress could be done mistakes in alignments accidentally when sewing shirts. If occurred the identified problem, the end product will be produced with minor quality. In that respect, need to provide an acceptable solution for the mentioned problem.

B. Proposed Solution

As a solution for the previously mentioned problem, When sewing men's shirts the seamstress should be kept 2" gap from the top button to the next nearest button. If the seamstress is able to keep the above-mentioned condition, then sewing defects can be minimized and capable to give an output to the better qualitative product. When providing a solution, a few conditions will be applied such as, shirts should be men's shirts and it should be plain color collared shirts (White color will be acceptable). To measure the button alignments of the shirt, need to find a method to detect the button as well as to check the button alignment with the gap of 2".[3] A condition will be applied such as the buttons should be a circular object. If the gap of 2 inches exceeds or reduce the sewing person can get an alert. Due to the mentioned process, the industry is able to reduce sewing defects and errors of productivity and that will be led until the quality product comes out.

III. Our Approach

A. Approach 1

When considering the button detection, at the first, needs to identify the button (a circular object) before checking the alignment. [4] As per the process, the below steps should be followed,

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- **First Step**
  The camera should be set to a fixed location. Seamstress should place the shirt (object) with facing the camera.
- **Second Step**
  The image will be appearing on the camera accordingly. The camera will be detecting the button circle through the tool of Hough circle which is able to detect round figures' and Determine the radius ranges of circle.
- **Third Step**
  Next attempt to find circles, the background should be bright (White shirts are preferable) and the buttons should be darker than the background (shirt) color. Therefore, need to be set the parameter 'Object Polarity' to 'Dark' for search dark circles.
- **Fourth Step**
  Detection sensitivity can be increased with doing the changes to pixel values. Draw the circles on the image. The circle centers will be positioning correctly.
- **Sixth Step**
  Find bright circles in the image or video. The light circles are almost the same intensity, maybe even brighter, as compared to the background. Therefore, to detect the light circle, need to change 'Object Polarity' to 'bright'. [5]
- **Seventh Step**
  Draw bright circles with different colors. Lower the value of the edge threshold and the distance by doing the required changes to the pixel values.

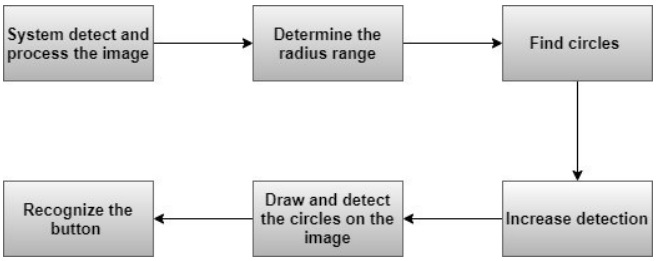

Figure 1: Button detection process

B. *Approach 2*

The next approach is to check the button alignment of the shirt. The condition will be applied as the system will detect the distance between the top button and the next nearest button to the top button. To do this, it is possible to apply fix range such that the gap between these two should be kept as 2 inches and that is a must. Therefore, need to be measured 2 inches between two buttons. [6]

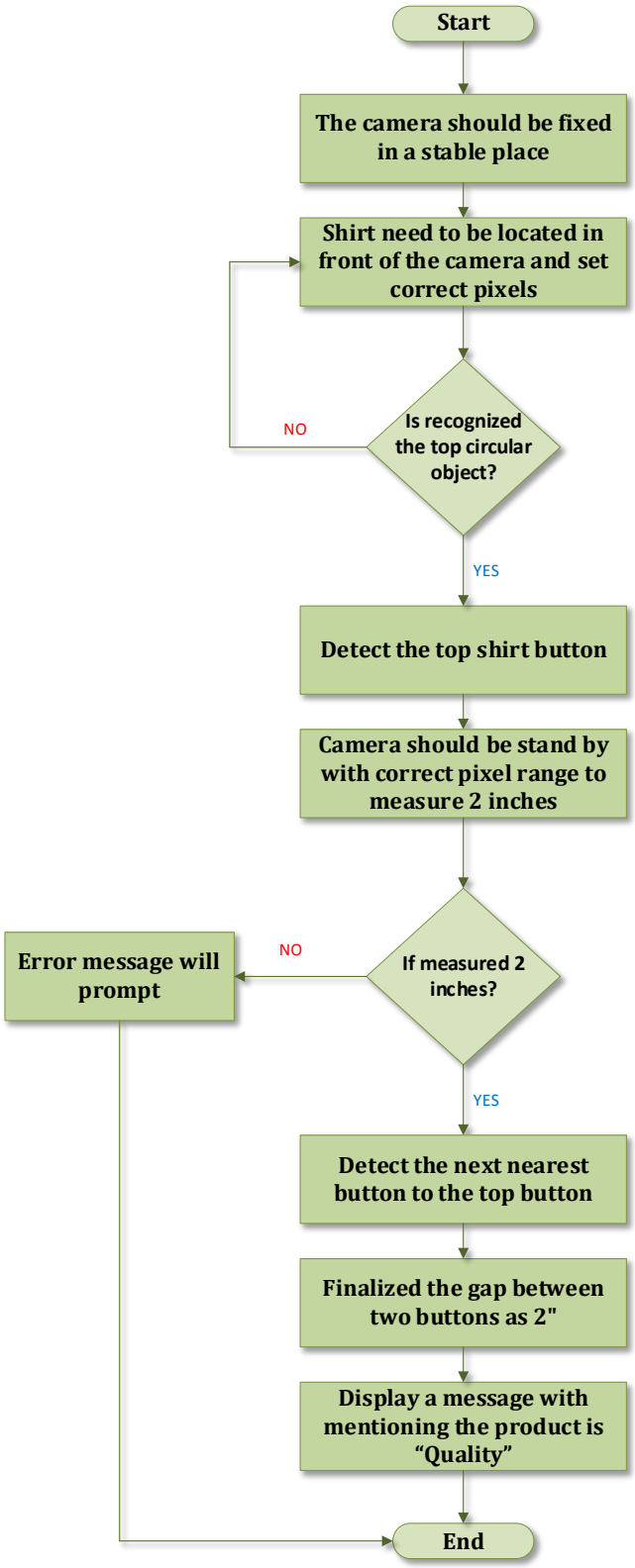

Figure 2: Button detection and alignment process

IV. Analyze and Design

The system is analyzed and designed with the use of appropriate technologies. As per the first approach of button detection, able to use Hough circles to detect and recognize the circle objects. (Buttons) As per the limitation, the system

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can recognize only circular buttons. [7] Basically, the Hough transform can be used to detect circular shapes.

A. *Analyzation of the Process Flow of Button Detection*

Step 1: Load image.
Step 2: Determine radius ranges for search circles.
Step 3: Initial attempt to find circles.
Step 4: Increase detection sensitivity.
Step 5: Draw the circle on the image.
Step 6: Find bright circles in the image.
Step 7: Draw bright circles with different color.
Step 8: Lower the value of head threshold.
Step 9: Draw dark and bright circles together.

After done the above process, the circular-shaped buttons can be detected and according to the mathematical logic, the positioning will be recognized by the system.

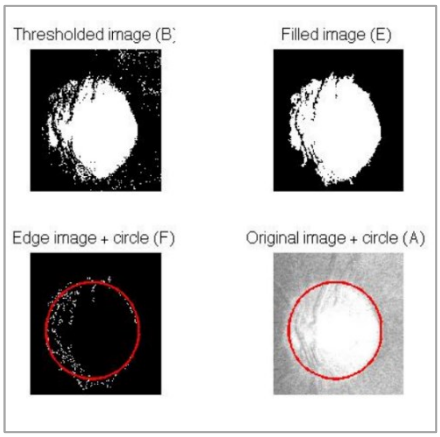

Figure 3: Detect Circular Objects (Buttons)

B. *Process Flow of Button Alignment*

Step 1: Detect the top button (Button near to the collar).
Step 2: Recognize the top button appropriately.
Step 3: Measure 2 inches from the top button to the next.
Step 4: Detect the next nearest button (second button).
Step 5: Recognize the second button appropriately.
Step 6: Proves that the system detects the quality product.
Step 7: Display an alert message.

C. *Units and Equations*

- Approach 1

To detect the circular objects (Buttons) the equation is as follow,

$$(x - x_{center})^2 + (y - y_{center})^2 = r^2$$

Equation 1

Where,

(x_{center}, y_{center}) is the center of the circle, and r is the radius of the circle. Under the above equation, it emphasizes that three parameters are available to detect the circular objects. Therefore, we need a 3D accumulator for Hough transform, which would be highly effective. [8] OpenCV provides more usability and maintainability to do the above task.

- Approach 2

To measure the button alignments between two buttons, the system should be measured 2 inches of gap between two buttons respectively. The camera will be able to position at a place, parallel to the sewing area. (Machine located area) The camera distance can be adjusted with do the required changes of pixels. [9]

V. Technology Adopted

A. Programming Language

The programming language used for the development of the system is Python and Mat lab. The main reason for using Python, because it is a high-level programming language that supports both object-oriented programming, functional programming and there's a high level of library support. Python has a design philosophy that emphasizes code readability, notably using significant whitespace. It provides constructs that enable clear programming on both small and large scales Python features a dynamic type system and automatic memory management. OpenCV Library was used along with Python for Image Cloning Detection Implementation. Blob detection is the used algorithm. [10]

B. Development Tools

PyCharm IDE was used as the primary IDEs for our project. Visual Studio and OpenCV have been used upon the necessity for developing the system. OpenCV is a library of programming functions that supports computer vision and that was the main library were used for this project. The main reason for using that library is because of the availability of a large number of functions and simplicity in practical use. The algorithms which used blob detection and Hough circles have been used to detect buttons easily.

VI. Implementation

A. *Approach 1 (Button Detection Process)*

This emphasizes the development of the button detection process. With respect to that, able to detect the top button of the shirt with the use of relevant technologies adopted to implement this project.

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To detect a circular type of object (button), follow the steps which have been already mentioned in Analyze and Design section and the technology is “Hough circles.” [11] Further, this particular section includes the code generation for the development process of Approach 1

Figure 4: View of Button Detection (Recognition)

B. *Approach 2 (Button Alignment Process)*

This is implemented after detecting the shirt buttons. As mentioned in Approach 1, at first have to detect the top button of the shirt. After find the top collared button, have to keep 2 inches’ distance and able to detect the second button. If there were any errors such as the gap between two buttons are less or greater than 2” the system will show it as an error. [12]

- ✓ If the button alignment increases or decreases 2 inches, the system is able to validate the status as “Accepted: No” and displays a validation message as “Rejected Product.”

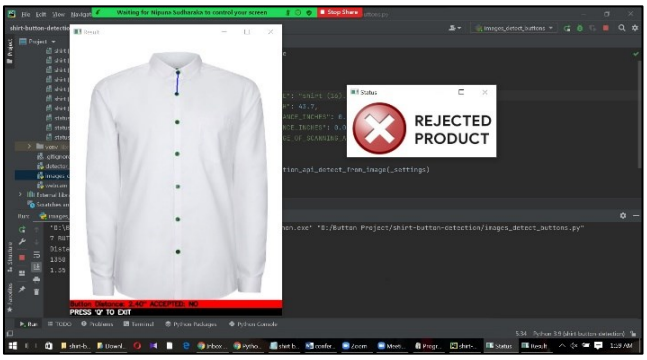

Figure 5: Validation for Rejected Product (Image Processing)

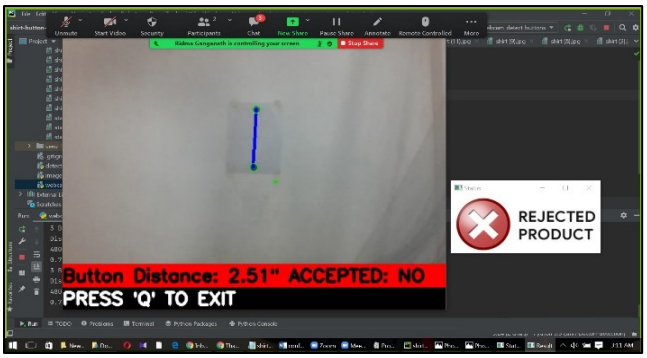

Figure 6: Validation for Increase of 2” (Video Processing)

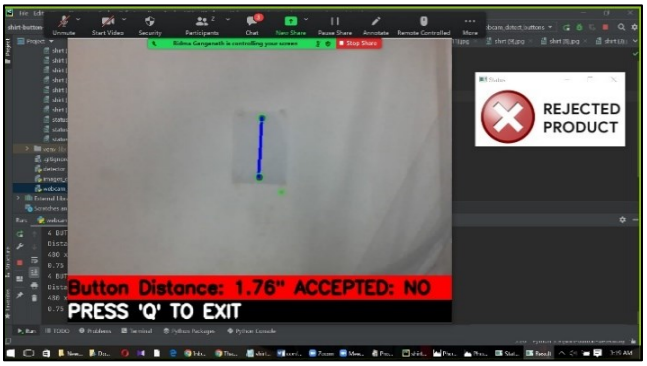

Figure 7: Validation for Decrease of 2” (Video Processing)

- ✓ If the button alignment is at most near to the 2 inches, the system able to validate the status as “Accepted: Yes” because either this can be acceptable or not acceptable and displays a validation message as “Not Up to Quality.”

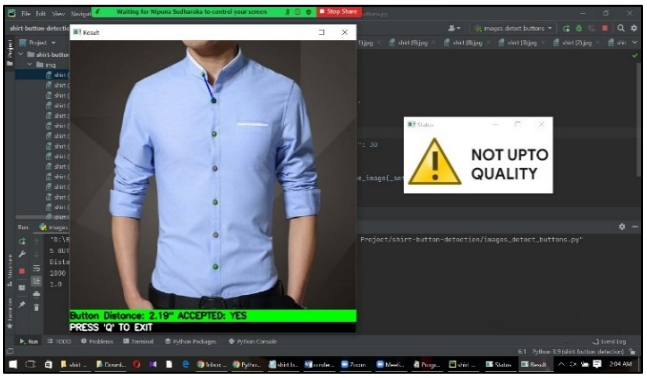

Figure 8: Validation for Not Up to Quality (Image Processing)

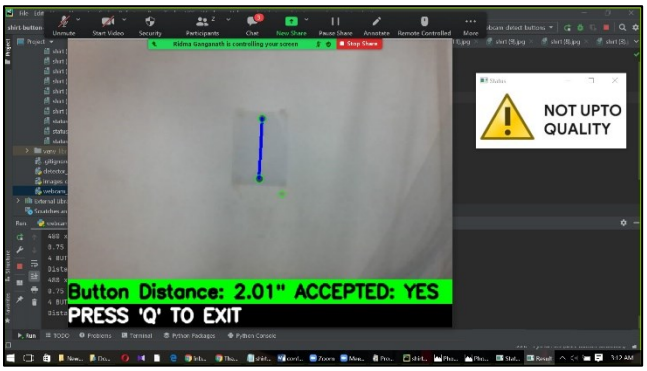

Figure 9: Validation for Not Up to Quality-Increase 2” (Video Processing)

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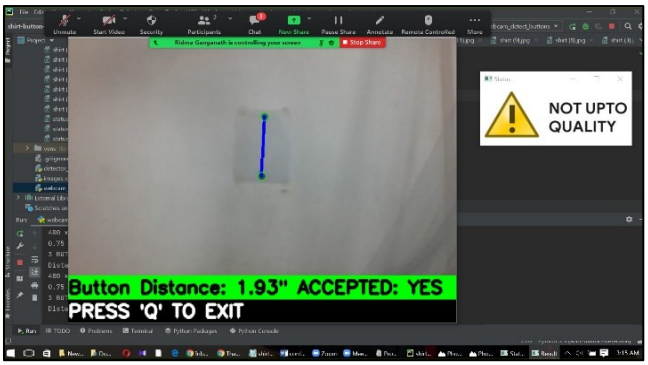

Figure 10: Validation for Not Up to Quality-Increase 2" (Video Processing)

- ✓ If the button alignment is equal to 2 inches, the system is able to validate the status as "Accepted: Yes" and displays a validation message as "Quality Product."

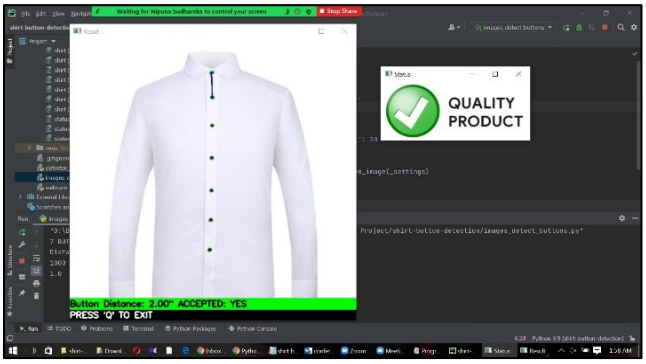

Figure 11: Validation for Quality Product (Image Processing)

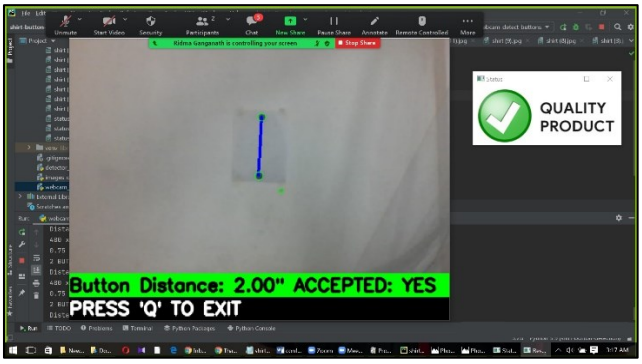

Figure 12: Validation for Quality Product (Video Processing)

VII. Evaluation

In this paper, the authors presented the results of the critical evaluation carried out on the benefits and limitations of method to check button alignment use in apparel industry. Presently there are some apparel systems available in the market and more are continuing to be developed by the industry and researchers alike. Every technology has its own advantages and disadvantages. Some systems are suitable for solving the problems in industry while others can be effectively perform. Some technologies perform better when there are only a few peers while others maintain their quality of operations even when large number of peers join in. Hence, it is impossible to identify one single technology that is suitable for all the requirements of every application and in the entire system. As future work, the authors propose to carry out a more in depth analysis with the objective coming out with a comprehensive taxonomy of textile. And will be able to find defects and detect errors more efficiently than this work.

VIII. Conclusion

The research came up with gaining solution and implementing a system to button detection and measure alignment. The system comprises of real time button detection and alignment. So far, have referred to the available literature covering respect to the relevant study areas and able to find the calculations used and approaches which were applicable for the research method.

Then the initial step carried out was analyzing the overall system and each module in the system in terms of input, process and output. The input of each module was identified as an image or video and the output is the detection result that defines whether an image or video is forged using a particular forging method or not. To design the process, first identified the procedures needed to be followed. There we have to customize the available calculations to suit our requirements and create new calculations to improve the process in terms of efficiency and accuracy as most of the available methods require high computational cost. Then we defined the objectives that need to be carried out for the successful completion of the system.

Due to the technical competencies, different technologies were selected. The main challenges that have been faced so far, understanding the available literature in terms of calculations, mathematical explanations, followed processes and etc. Sometimes we were unable to find relevant datasets from other researchers where we had to implement our own datasets.

As the future work, it is a must to improve the performance and the accuracy of these calculations, algorithms in order to cover a wide range of images and to detect wide range of videos which process in real time using NVIDIA Jetson Nano. There are restrictions on input images from which we try to cover the pitfalls in these algorithms. But it would be really helpful if these algorithms are in a state where they are 100% capable to detect errors. Once the performance and the accuracy increased, this system will be a trending one among people because of the rising necessities and to continue this research to a great height.

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